FWA-01 / HEALTHCARE FWA KNOWLEDGE BASE

Claims Integrity

Claims validation, payment accuracy and PRV00092

Scope and synthetic policy

Purpose and boundaries

This fictional payer knowledge base supports a RAG training exercise about healthcare Fraud, Waste and Abuse (FWA). All organizations, identifiers, periods, financial amounts, rules and outcomes are invented. DOH means a generic department of health in this exercise; no content represents any actual DOH policy, dataset or finding. An anomaly is a reason to review evidence, not proof of fraud. Monetary values use synthetic currency units (SCU).

CI-P01 | Coverage and documentation

For this exercise, a payable claim must have an active synthetic member on the service date, a valid provider record, an eligible benefit and documentation supporting the billed service. Prior authorization is required only for services listed in the fictional benefit configuration. Reviewers must apply the policy version effective on the service date. A missing authorization may be an administrative defect rather than intentional deception.

CI-P02 | Duplicate and coding review

An exact-duplicate candidate shares member_id, provider_id, service_date, procedure_code, units and billed_amount with another non-void line. A corrected claim linked to its original must be evaluated as a replacement, not counted twice. A near-duplicate has similar service facts but differing modifiers, units or amounts; it requires human review. Upcoding means billing a higher-intensity service than the evidence supports; statistical intensity alone does not establish upcoding.

CI-P03 | Fair decision process

The analyst records the triggering rule, claim lineage, policy version and supporting documents. The provider receives a request for clarification before a final adverse determination in this fictional workflow. A reviewer independent of the original decision handles reconsideration. Never describe a flagged amount as confirmed loss or savings. These are training controls, not legal requirements.

RAG learning objective

Begin with document loading, page-level text extraction and source metadata. Keep CI-P01, CI-P02 and CI-P03 intact where possible so a retrieved answer includes the rule and its exceptions. The assistant must identify this document as synthetic when discussing its policies.

Operational workflow and evidence

Step 1 | Receive and normalize

Ingest claim headers and lines separately. Retain claim_id, line_id, original_claim_id, claim_status, service_date, submission_date, member_id, provider_id, procedure_code, modifiers, units, billed_amount and paid_amount. Standardize dates and code casing without changing the original values. Quarantine negative units, missing keys and impossible dates; store the rejection reason.

Step 2 | Validate before detecting

Join to provider and member reference records using date-effective keys. Distinguish submitted, rejected, accepted, adjudicated and paid lines. Exclude void lines from active duplicate detection and explicitly reconcile replacements. Check that ingestion has not duplicated a source file before treating repeated rows as provider behavior.

Step 3 | Generate and prioritize candidates

Run exact-duplicate, overlapping-service, excessive-unit and coding-consistency rules. Give every finding an alert_id and rule_version. Collapse multiple alerts on one claim into a case while retaining individual reasons. Rank cases by estimated exposure, clinical relevance, confidence and reviewer capacity. The priority score is a work queue aid, not a determination.

Step 4 | Review, resolve and learn

A reviewer compares source claims, visit records and fictional policy. Record one of: supported, administrative correction, insufficient evidence, confirmed overpayment or escalated investigation. Finance verifies an actual recovery before it enters the recovered-amount metric. Feed adjudicated outcomes to evaluation data only after checking label quality and possible reviewer disagreement.

Evidence and access

Store a reproducible snapshot of the claim lines, provider reference version, rule parameters and reviewer notes. Use pseudonymous member keys in the exercise. Separate claim facts from free-text allegations. Limit analyst access by role and keep an audit event for evidence access and case changes. An answer must cite the policy page and distinguish a claim fact from an analyst hypothesis.

Illustrative PRV00092 metrics

Measurement frame

PRV00092 is a wholly fictional provider. The illustrative observation window is 1-30 September 2026. There are 12,000 accepted claim lines, 8,400 unique encounters and 3,100 unique synthetic members. Paid amount is 2,400,000 SCU. The dataset is a fixed teaching snapshot, not a live or externally verified report.

Metric table

Measure	Synthetic value / definition
Exact-duplicate candidates	240 / 12,000 accepted lines = 2.0%
Near-duplicate candidates	360 / 12,000 accepted lines = 3.0%
All candidate lines	540 unique lines; 60 overlap both rules
Reviewed candidate lines	180 of 540 = 33.3% review coverage
Confirmed overpayment lines	45 of 180 reviewed = 25.0% review yield
Confirmed overpayment amount	18,000 SCU; limited to reviewed cases
Verified recovered amount	12,000 SCU; 66.7% of confirmed amount
Candidate paid exposure	90,000 SCU on 540 unique lines

Interpretation and denominator discipline

Candidate rate is 540 / 12,000 = 4.5%, not 5.0%, because overlapping rules are deduplicated. The 25.0% review yield cannot be treated as a population fraud rate: the reviewed sample was prioritized, not random. Candidate paid exposure is the paid value of flagged lines and is not expected recovery. Do not add 90,000, 18,000 and 12,000; they describe different stages of the same possible exposure.

Comparison example

A fictional specialty-and-volume peer group has a 1.2% median exact-duplicate candidate rate across 20 providers. PRV00092 is 0.8 percentage points above that median. This is not a risk-adjusted causal comparison. Check specialty mix, source refreshes and replacement-claim handling before drawing conclusions. Small counts, seasonality and rule changes can distort month-to-month trends.

RAG test

Question: What share of PRV00092 lines were unique candidates, and how much was recovered? Expected answer: 4.5% candidate rate and 12,000 SCU verified recovery; cite this page and say both are synthetic. A response reporting 90,000 SCU recovered fails the test.

Case scenarios and interview practice

CI-S01 | Replacement mistaken for duplicate

Two lines for member SYN-M014 show the same procedure on the same date. The second claim refers to the first through original_claim_id and the first is void. Expected disposition: exclude the void line and assess the active replacement. Evidence needed: status history, linkage and payment ledger. Lesson: a matching key is an alert feature, not a final payment decision.

CI-S02 | Unsupported units

PRV00092 bills four therapy units while a synthetic attendance record supports two. A reviewer requests the session note and checks the fictional unit convention. If the provider supplies a valid correction, record an administrative adjustment; if the final evidence supports only two units, calculate the overpayment from the allowed paid unit rate. Do not infer intent from this example.

CI-S03 | Missing evidence

A claim is flagged for high service intensity but the clinical record is unavailable. Expected disposition: insufficient evidence with a pending request. The RAG assistant should explain the missing evidence, not invent clinical details or declare fraud. A failed document retrieval should remain distinguishable from an actual absence of documentation.

Hands-on tasks

Load this PDF, preserve page numbers and build a baseline vector index. Ask about duplicate exceptions and PRV00092 recovery. Compare fixed-size chunking with section-based chunking. Add a metadata filter for doc_id=FWA-01 and verify the answer changes when the correct policy page is excluded. Save retrieved chunks alongside each answer for inspection.

Interview prompts with answer anchors

Why deduplicate alerts? Several rules can flag the same line, so summing rule counts exaggerates exposure. Why is recovery different from detection? Recovery requires confirmed financial movement. Why use overlap in chunking? It can preserve context across boundaries but increases index size and repeated evidence. How would you avoid hallucinated amounts? Retrieve the metric definitions, cite the exact page and abstain when the answer is absent.

Cross-document links

Use FWA-02 for provider-level prioritization and fair comparisons. Use FWA-06 for ingestion, evaluation and the complete RAG learning path. This PDF is reference knowledge for the exercise; it is not a real reimbursement or investigation manual.